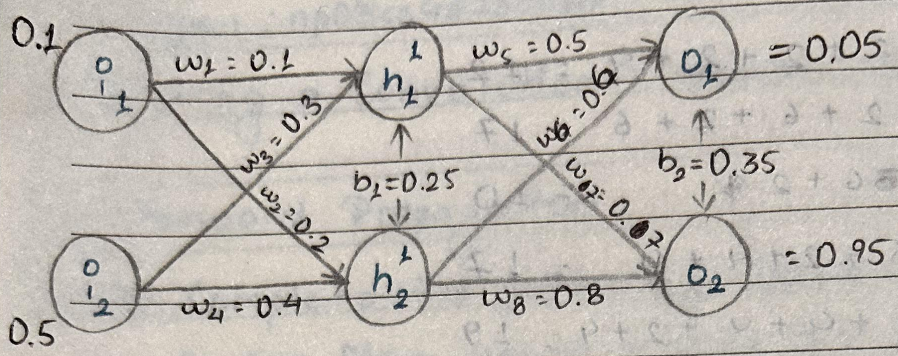
Tutorial #5 : Narayan Lohani ; np03cs4a230039



Using two inputs $[i_1 = 0.1 ; i_2 = 0.5]$

* Perform forward pass through the network & computer total error
* Perform backward pass to propagate the error within the network & update the weights.

→ Given,

$i_1 = 0.1$; $i_2 = 0.5$; $b_1 = 0.25$; $b_2 = 0.35$

$w_1 = 0.1$; $w_3 = 0.3$; $w_5 = 0.5$; $w_7 = 0.7$

$w_2 = 0.2$; $w_4 = 0.4$; $w_6 = 0.6$; $w_8 = 0.8$

$y_1 = 0.05$; $y_2 = 0.95$

Hidden layer neurons :

$h_1^1 = \cancel{i_1 \times w_1}\ \sigma(z_1^1)$ [sigmoid of z_1^1]

$z_1^1 = (i_1 \times w_1) + (i_2 \times w_3) + b_1$

$= (0.1 \times 0.1) + (0.5 \times 0.3) + 0.25$

$= 0.41$

$h_1^1 = \sigma(0.41) = 0.6010879$

$z_2^1 = (i_1 \times w_2) + (i_2 \times w_4) + b_1$

$= (0.1 \times 0.2) + (0.5 \times 0.4) + 0.25$

$= 0.47$

$h_2^1 = \sigma(z_2^1) = \sigma(0.47) = 0.6153838$

$zo_1 = (h_1^1 \times w_5) + (h_2^1 \times w_6) + b_2$

$= (0.60 \times 0.5) + (0.61 \times 0.6) + 0.35$

$= 1.01834464$

$o_1 = \sigma(zo_1) = \sigma(1.01) = 0.73465$

$zo_2 = (h_1^1 \times w_7) + (h_2^1 \times w_8) + b_2$

$= (0.60 \times 0.7) + (0.61 \times 0.8) + 0.35$

$= 1.26163898$

$o_2 = \sigma(zo_2) = \sigma(1.26) = 0.7793081$

Error Calculations:

$E_1 = \frac{1}{2} \Sigma (y_1 - o_1)^2$

$= \frac{1}{2} (0.05 - 0.73465)^2$

$= 0.23437281$

$$E_2 = \frac{1}{2} \Sigma (y_2 - O_2)^2$$

$$= \frac{1}{2} (0.95 - 0.77)^2$$

$$= 0.01456786$$

Total error(J) = $E_1 + E_2$

$= 0.23437281 + 0.01456786$

$= 0.24894067$

Calculating Gradients of Error w.r.t. weights:

$$\frac{dJ}{dw} \Rightarrow \left[\frac{\partial J}{\partial w_5}, \frac{\partial J}{\partial w_6}, \frac{\partial J}{\partial w_7}, \frac{\partial J}{\partial w_8}; \frac{\partial J}{\partial w_1}, \frac{\partial J}{\partial w_2}, \frac{\partial J}{\partial w_3}, \frac{\partial J}{\partial w_4} \right]$$

$$\frac{\partial J}{\partial w_5} = \frac{\partial J}{\partial y_1} \cdot \frac{\partial y_1}{\partial zo_1} \cdot \frac{\partial zo_1}{\partial w_5}$$

$$\frac{\partial J}{\partial y_1} = \frac{\partial \left(\frac{1}{2} \Sigma (\hat{y}_1 - \hat{y}_1)^2 \right)}{\partial y_1} \quad [\hat{y}_1 = O_1]$$

$$= \frac{1}{2} \cdot 2(\hat{y}_1 - y_1) \cdot 1$$

$$= \cancel{(y_1 - O_1)}\ (O_1 - y_1)$$

$$\frac{\partial y_1}{\partial zo_1} = \frac{\partial \sigma(zo_1)}{\partial zo_1} = o_1(1 - o_1)$$

$$\frac{\partial zo_1}{\partial w_5} = \frac{\partial(w_5 \times h_1^1 + w_6 \times h_2^1 + b_2)}{\partial w_5} = h_1^1$$

$$\frac{\partial J}{\partial w_5} = (\overset{O_1 - y_1}{\cancel{y_1 - O_1}}) \times \left[O_1(1 - O_1)\right] \times h_1^1$$

$$= (\cancel{0.05}\,0.73 - 0.05) \times \left[0.73 \times (1 - 0.73)\right] \times 0.60$$

(1) $= 0.08022434$

$$\frac{\partial J}{\partial w_6} = \frac{\partial J}{\partial y_1} \cdot \frac{\partial y_1}{\partial zo_1} \cdot \frac{\partial zo_1}{\partial w_6}$$

$\frac{\partial J}{\partial y_1}$ and $\frac{\partial y_1}{\partial zo_1}$ are calculated in $\frac{\partial J}{\partial w_5}$ and are same for w_6 too. so, calculating $\frac{\partial zo_1}{\partial w_6}$ only:

$$\frac{\partial zo_1}{\partial w_6} = \frac{\partial(w_5 \times h_1^1 + w_6 \times h_2^1 + b_2)}{\partial w_6} = h_2^1$$

$$\frac{\partial J}{\partial w_6} = \frac{\partial J}{\partial y_1} \cdot \frac{\partial y_1}{\partial zo_1} \cdot \frac{\partial zo_1}{\partial w_6}$$

$$= (\overset{O_1 - y_1}{\cancel{y_1 - O_1}}) \left[O_1 (1 - O_1) \right] \cdot h_2^1$$

$$= (0.73 - 0.05) \left[0.73 (1 - 0.73) \right] \cdot 0.61$$

(2) $= 0.08213235$

$$\frac{\partial J}{\partial w_7} = \frac{\partial J}{\partial y_2} \cdot \frac{\partial y_2}{\partial zo_2} \cdot \frac{\partial zo_2}{\partial w_7}$$

$$\frac{\partial J}{\partial y_2} = \frac{\partial \left(\frac{1}{2} \Sigma (\overset{O_2 - y_2}{\cancel{y_2 - O_2}})^2 \right)}{\partial y_2} = \cancel{y_2 - O_2}\ O_2 - y_2$$

$$\frac{\partial y_2}{\partial zo_2} = \frac{\partial \sigma(zo_2)}{\partial zo_2} = O_2 (1 - O_2)$$

$$\frac{\partial zo_2}{\partial w_7} = \frac{\partial (h_1' w_7 + h_2' w_8 + b_2)}{\partial w_7} = h_1'$$

$$\frac{\partial J}{\partial w_7} = (\overset{O_2 - y_2}{\cancel{y_2 - O_2}}) \left[O_2 (1 - O_2) \right] \cdot h_1'$$

$$= \left[0.77 - 0.95 \right] \left[0.77 (1 - 0.77) \right] . 0.60$$

③ $= -0.0191268$

$$\frac{\partial J}{\partial w_8} = \frac{\partial J}{\partial y_2} \cdot \frac{\partial y_2}{\partial zo_2} \cdot \frac{\partial zo_2}{\partial w_8}$$

$\frac{\partial J}{\partial y_2}$ and $\frac{\partial y_2}{\partial zo_2}$ are already calculated for w_7. so, only calculating dzo_2/dw_8.

$$\frac{\partial zo_2}{\partial w_8} = \frac{\partial(h_1' w_7 + h_2' w_8 + b_2)}{\partial w_8} = h_2'$$

$$\therefore \frac{\partial J}{\partial w_8} = \frac{\partial J}{\partial y_2} \cdot \frac{\partial y_2}{\partial zo_2} \cdot \frac{\partial zo_2}{\partial w_8}$$

$$= (O_2 - y_2)\left[O_2(1-O_2)\right] \cdot h_2'$$

$$= (0.77 - 0.95)\left[0.77(1-0.77)\right] \cdot 0.61$$

④ $= -0.01944558$

Weight update of second layer is in next page. (page-8)

Now, for weights of first layer

$$\frac{\partial J}{\partial w_1} = \frac{\partial J}{\partial y_2} \cdot \frac{\partial y_2}{\partial zo_2} \cdot \frac{\partial zo_2}{\partial h_1'} \cdot \frac{\partial h_1'}{\partial z_1'} \cdot \frac{\partial z_1'}{\partial w_1}$$

$$\cancel{\frac{\partial J}{\partial w_1} = \frac{\partial J}{\partial y_1} \cdot \partial y_1 \frac{\partial J}{\partial h_1^1} \cdot \frac{\partial h_1^1}{\partial z_1^1} \cdot \frac{\partial z_1^1}{\partial w_1}}$$

$$\frac{\partial J}{\partial w_1} = \frac{\partial J}{\partial o_1} \cdot \frac{\partial o_1}{\partial zo_1} \cdot \frac{\partial zo_1}{\partial h_1^1} \cdot \frac{\partial h_1^1}{\partial z_1^1} \cdot \frac{\partial z_1^1}{\partial w_1} \left[\cancel{o_1 - y} \right.$$

$$\frac{\partial J}{\partial o_1} = \frac{\partial \frac{1}{2}\left(\Sigma (o_1 - y_1)^2\right)}{\partial o_1} = o_1 - y_1$$

$$\frac{\partial o_1}{\partial zo_1} = \frac{\partial \sigma(zo_1)}{\partial zo_1} = o_1(1 - o_1)$$

$$\frac{\partial zo_1}{\partial h_1^1} = \frac{\partial \left[h_1^1 . w_5 + h_2^1 . w_6 + b \right]}{\partial h_1^1} = w_5$$

$$\frac{\partial h_1^1}{\partial z_1^1} = \frac{\partial \sigma(z_1^1)}{\partial z_1^1} = h_1^1(1 - h_1^1)$$

$$\frac{\partial z_1^1}{\partial w_1} = \frac{\partial (i_1 w_1 + i_2 w_3 + b)}{\partial w_1} = i_1$$

Combining everything:

$$\frac{\partial J}{\partial w_1} = (o_1 - y_1) . o_1 (1 - o_1) . w_5 . h_1^1 (1 - h_1^1) . i_1$$

$$= (0.73 - 0.05)\, 0.73 (1 - 0.73) . 0.4919\, 0.60 (1 - 0.60) \times 0.1$$

5 $= 0.001582280$

Updated weights of second layer.

$\eta = 0.1$

$$w_5 = w_5 - \eta . \frac{\partial J}{\partial w_5} = 0.5 - 0.1(0.080) = 0.4919$$

$$w_6 = w_6 - \eta \frac{\partial J}{\partial w_6} = 0.6 - 0.1(0.082) = 0.5917$$

$$w_7 = w_7 - \eta \frac{\partial J}{\partial w_7} = 0.7 - 0.1(-0.019) = 0.7019$$

$$w_8 = w_8 - \eta \frac{\partial J}{\partial w_8} = 0.8 - 0.1(-0.019) = 0.8019$$

$$\frac{\partial J}{\partial w_2} = \frac{\partial J}{\partial o_2} \cdot \frac{\partial o_2}{\partial zo_2} \cdot \frac{\partial zo_2}{\partial h'_2} \cdot \frac{\partial h'_2}{\partial z'_2} \cdot \frac{\partial z'_2}{\partial w_2}$$

$$\frac{\partial J}{\partial o_2} = \frac{\partial\left(\frac{1}{2}\left(\Sigma(o_2 - y_2)^2\right)\right)}{\partial o_2} = o_2 - y_2$$

$$\frac{\partial o_2}{\partial zo_2} = \frac{\partial \sigma(zo_2)}{\partial zo_2} = o_2(1 - o_2)$$

$$\frac{\partial zo_2}{\partial h'_2} = \frac{\partial(h'_1 . w_7 + h'_2 . w_8 + b)}{\partial h'_2} = w_8$$

$$\frac{\partial h'_2}{\partial z'_2} = \frac{\partial \sigma(z'_2)}{\partial z'_2} = h'_2(1 - h'_2)$$

$$\frac{\partial z'_2}{\partial w_2} = \frac{\partial(i_1 . w_2 + i_2 . w_4 + b)}{\partial w_2} = i_1$$

$$\frac{\partial J}{\partial w_2} = (o_2 - y_2)\, o_2(1 - o_2)\, w_8 . h'_2(1 - h'_2) . i_1$$

$$= (0.77 - 0.95)\, 0.77(1 - 0.77) . 0.8019 . 0.61(1 - 0.61) . 0.1$$

6 $= -0.000608143$

$$\frac{\partial J}{\partial w_3} = \frac{\partial J}{\partial o_1} \cdot \frac{\partial o_1}{\partial zo_1} \cdot \frac{\partial zo_1}{\partial h'_1} \cdot \frac{\partial h'_1}{\partial z'_1} \cdot \frac{\partial z'_1}{\partial w_3}$$

$$\frac{\partial z'_1}{\partial w_3} = \frac{\partial(i_1.w_1 + i_2.w_3 + b)}{\partial w_3} = i_2$$

$$\frac{\partial J}{\partial w_3} = (o_1 - y_1).o_1.(1-o_1).w_5.h'_1(1-h'_1).i_2$$

$$= (0.73 - 0.05)\,0.73\,(1-0.73).0.4919.0.60(1-0.60).0.5$$

7 $= 0.00791140$

$$\frac{\partial J}{\partial w_4} = \frac{\partial J}{\partial o_2} \cdot \frac{\partial o_2}{\partial zo_2} \cdot \frac{\partial zo_2}{\partial h'_2} \cdot \frac{\partial h'_2}{\partial z'_2} \cdot \frac{\partial z'_2}{\partial w_4}$$

$$\frac{\partial z'_2}{\partial w_4} = \frac{\partial(i_1.w_2 + i_2.w_4 + b)}{\partial w_4} = i_2$$

$$\frac{\partial J}{\partial w_4} = (o_2 - y_2).o_2(1-o_2).w_8.h'_2(1-h'_2).i_2$$

$$= (0.77 - 0.95).0.77(1-0.77) \cdot 0.8019 \cdot 0.61(1-0.61).0.5$$

8 $= -0.003040715$

Updated weights of first layer.

$\eta = 0.1$

$$w_1 = w_1 - \eta \frac{\partial J}{\partial w_1} = 0.1 - 0.1(0.0015) = 0.09985$$

$$w_2 = w_2 - \eta \frac{\partial J}{\partial w_2} = 0.2 - 0.1(-0.00060) = 0.20006$$

$$w_3 = w_3 - \eta \frac{\partial J}{\partial w_3} = 0.3 - 0.1(0.00791) = 0.299209$$

$$w_4 = w_4 - \eta \frac{\partial J}{\partial w_4} = 0.4 - 0.1(-0.00304) = 0.400304$$